



REVIEW ARTICLE

Assessment of the effect of site-specific nutrient management on rice crop in South Asia: A review

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Abstract

Nitrogen (N) is an essential mineral nutrient for plant growth and the demand for nitrogenous fertilisers has increased significantly to achieve higher grain production per unit area. However, excessive application of N can lead to serious environmental problems, particularly water pollution (through nitrate leaching and runoff) and air pollution (through emissions of ammonia and nitrous oxide). To reduce the negative environmental impact of N while ensuring sustainable crop production, efficient N management is essential. The present study comprises N management site-specific nutrient management (SSNM) tools such as leaf colour chart (LCC), soil plant analyser development (SPAD) and rice crop manager (RCM) and certain strategies that have been introduced in agriculture to optimise fertiliser application rates, minimise N losses and supply adequate nutrients to crops according to their demand. Proper N management not only improves nutrient use efficiency but also enhances productivity and environmental sustainability. The SSNM is recognised as proven technology for the better management of N. We collected research papers published in reputed international journals on SSNM and precise N management during 2000–2025. The relevant data from the research papers was systematically tabulated and univariate analysis was performed to draw the interferences. The yield increment was 8.14 % in SSNM experiments over blanket application of N fertiliser. Nitrogen dose was drastically reduced by the application of N as per SSNM principle; it reduces around 12 kg N ha⁻¹, which is 9.38 % lower N than the non-SSNM treatment. The partial factor efficiency, which is around 22.31 % higher in SSNM applied field than control treatment. While the agronomic efficiency was varied from 12.84–17.58 in farmers fertilisers practice (FFP) and SSNM, respectively, this is 36.9 % higher in SSNM than control treatment. Therefore, SSNM intervention needs to be popularised among the farmer's for environmental sustainability and food security.

Keywords: leaf colour chart; nitrogen; rice; rice crop manager; SSNM

Introduction

One alternative to a suggestion for consistent fertiliser application over a rice production system inside a nation or area is site-specific nutrient management (SSNM). In general consumption of the fertiliser is more in cereals as compare to other crops. The most of south Asian soils particularly Indian soils are deficient in available nitrogen (N) (1). However, N present in soil cannot fulfil demand of rice crop during the critical growth stages (2). It is very important to match the optimal dose of nitrogenous fertiliser for improving the crop growth and development (3). Nitrogenous fertiliser affects the quick response of the crop in terms of the growth of plants, particularly urea. Excessive use of nitrogenous fertilisers adversely degrades the environment, particularly air and water (4). Nitrate (NO₃) content has increased in surface water bodies as well as ground water which are harmful for health purpose (5).

According to World Health Organization (WHO), NO₃⁻ content exceeding more than 10 mg is not suitable for drinking purposes (6). Thus, inputs of N fertiliser in agricultural land can be a key reason in

eutrophication (“improving nutrition”). Due to eutrophication, too much growth of algae and macrophytes cause oxygen depletion, this unavailability of oxygen is toxic to aquatic organisms (7).

Improper application of fertilisers creates global warming problems which elevated the serious concern about the existing nutrient management practices. Denitrification is the process of microbial conversion of the NO₃⁻ into nitrous oxide (N₂O) produces and accounts for approximately 5 % of the total greenhouse gas (CHG) (8). Nitrous oxide is also responsible for the depletion of the ozone layer. The N₂O is also necessary for the atmosphere because it has maximum lifespan than other gases and the concentration is increased by 0.22 % per year (9). It is essential to manage the application of N fertiliser in farmers field.

Proper nutrient management through SSNM is vital to improve the productivity of the crops. Nitrogenous fertilisers are costly inputs and it is estimated that in agriculture more than \$45 billion per year is invested on nitrogenous fertilisers (6). Commonly, more than 50 % of the applied N is not utilised by the plants (10, 11).

It is proven in many studies that N is a limiting nutrient, around 1 kg of N produces around 60 kg of rice (12). The demand of N was less before green revolution but after green revolution, the quick response of N over the crops makes it the most demanded fertiliser when compared to other fertilisers (13). Nitrogenous fertilisers can boost food production of wheat about 260 % (average growth of about 6.4 % per year). Presently for food production, half of the world's population directly depends upon N containing fertilisers (6). Indian soils are deficient in organic N content, hence to fulfil the requirement of N to crops the farmer's are depends on the chemical nitrogenous fertiliser. It is very challenging to sustainably manage the N management for better crop production. SSNM is a technique to supply the essential nutrients optimally as and when crop is needed. It was assessed that the yield of rice in Asia increase about 14 % during 2000–2010 and by 25 % from 2000–2020. However, it was supposed that the average yields of irrigated rice must rise from 5.3 t ha⁻¹ in 1998 to 6.5–7.0 t ha⁻¹ in 2020 (14). In this review our aim is to evaluate the impact of SSNM tools like Leaf Colour Chart (LCC) and Rice Crop Manager (RCM) on N footprint on environment and nutrient use efficiency indices like agronomic use efficiency and partial factor productivity.

Methodology

The systematic research papers on the site-specific nutrient management were selected for the study. The period considered for the study was 2000–2025, 78 research papers were collected from Google scholar and 40 among them were finalised for data extraction which was based on all over the world field experiments. All the collected data from the research paper were summarised on Microsoft Excel sheet. The data was extracted as emission of greenhouse gases (GHGs) in response of SSNM, yields of rice and efficiency indices (agronomic efficiency, recovery efficiency and partial factor productivity). The formula of determination of different indices are given below:

$$\text{Agronomic efficiency} = \frac{(\text{Grain yield in fertilised plot} - \text{Grain yield in omission plot})}{(\text{Fertiliser N applied})} \times 100 \quad \text{Eqn. 1 (6)}$$

$$\text{Partial factor productivity} = \frac{\text{Grain yield in fertilised plot}}{\text{Fertiliser N applied}} \quad \text{Eqn. 2 (6)}$$

$$\text{Recovery efficiency} = \frac{(\text{N uptake in fertilised plot} - \text{N uptake in omission plot})}{\text{Fertiliser N applied}} \quad \text{Eqn. 3 (6)}$$

Tabulated data was subjected to univariate analysis. The standard error of mean was calculated to evaluate the variability in present data by using online software statistical tools for agriculture analysis (STAR Nebula).

Site-specific nutrient management tools for nutrient management in rice crop

Leaf colour chart (LCC)

LCC is an easy, low-cost way to use colour to determine a crop's N needs. There were six green LCC strips displayed, ranging in tint from light green to dark green as shown in Fig. 1. The presence of N content in the leaf can be determined by evaluating its greenness using these colour strips. The LCC reading of the standing rice crop can be started at peak N requirement stages that is starting of tillers to panicle formation stage, LCC is used to take readings in rice production fields. Research on nitrogenous fertiliser based on the LCC showed that it saved 51, 20 and 29 kg ha⁻¹ of nitrogenous fertiliser compared to blanket N fertiliser, while producing same or better grain production (15). Fertiliser N management using inexpensive SSNM technologies like rice crop manager and LCC can reduce the extreme application of N by 10–50 % (14). This is because it is inexpensive, simple to use and provides a visual comparison of the greenness of crop leaves. The LCC is easily within reach of most farmer's.

Under conditions of varying temperatures, planting times and other agronomic practices, N dosages can be modified in accordance with the readings of the leaf chlorophyll content chart, which measures the amount of green content in the leaf (16). Different leaf temperatures (15–35 °C) can cause SPAD values to fluctuate, even when the physiological chlorophyll level is the same. This indicates that temperature affects the accuracy of measurements. Ideally in the morning or late afternoon when the direct heat stress is at its lowest, take SPAD/LCC readings in a stable temperature environment and stay away from readings during the midday heat (17). SPAD readings are typically collected every 7–10 days from the early vegetative phase through the time of panicle initiation, which is close to the reproductive phase. From the start of tillering (or early leaf growth) until after reproductive stage (e.g., panicle emergence in rice), LCC measurements should ideally be obtained every week. Meanwhile, there was a strong correlation between the N content in leaf tissue and chlorophyll index values. The chlorophyll molecule contains N in plants as a necessary



Fig. 1. Picture of International Rice Research Institute leaf colour chart and Punjab Agriculture University leaf colour chart guided nitrogen application.

element. Nitrogen content in plant was found in the porphyrin ring structure of every chlorophyll molecule. Sufficient N boosts chlorophyll production in plants. The chlorophyll content was determined by devices like SPAD meters, which measure light absorption in leaves. Chlorophyll index values are an indirect indicator of the N content of leaves because chlorophyll levels rise with N availability. Between leaf N concentration and SPAD chlorophyll measurements, the agronomic trials revealed strong positive correlations ($r=0.6-0.9$) (18–20). Because of its user-friendliness, adaptability to different climates, affordability and ease of use, LCC is a tool that farmer's can rely on.

Use of the International Rice Research Institute (IRRI) leaf colour chart (LCC) : In the first step, 14 days after transplanting (DAT) select randomly 10 plants across the field where they are uniformly distributed, healthy, fully expanded then topmost leaf of plant is compare from each of the 10 plants with the LCC. The leaf is placed on the middle part of strips for comparison and do not remove the leaf from the plants. Readings are taken in the morning time because in direct sunlight readings cannot be easily noted down. If the reading of 6 or 10 leaves of the plant is below the critical value 4, N fertiliser are needed to be applied to the crop. According to LCC critical value, applied N fertiliser dose is different for basmati and non-basmati rice.

Wet season (Kharif) 28 kg ha⁻¹ urea is applied to non-basmati rice at LCC value <4.

Wet season (Kharif) 23 kg ha⁻¹ urea is applied to basmati rice at LCC value <3.

Use of the Punjab Agriculture University (PAU) leaf colour chart (LCC): Select only representative plants and match colour of the intact leaves with the LCC under the shade. The plants selected for matching colour was normal and free from disease/insect incidence, water logging/stress other nutrient deficiencies. In transplanted rice, 25 kg urea is required to be applied acre⁻¹. Compare the colour of leaf to the LCC at 7 days interval, after 14 days of transplanting. Whenever, the greenness of 6 or more out of 10 leaves is lighter than LCC shade 4, apply 25 kg urea per acre. If leaf colour shade is darker than LCC value 4 then urea should not be applied in transplanted rice. After the initiation of flowers, discontinue the application of urea. The LCC reading is taken from the normal, disease/insect incidence free and nutrient deficient should not be visible on plants leave.

Rice crop manager (RCM)

Presently in this digital revolution, everyone is using smartphones to get rapid results for any query. The nutrient management of rice crop is not only depended upon the soil type but also depends upon the crop cultivation practices, duration of cultivars and other management factors, which varies among fields within a rice-growing area. Rice crop manager is a web-based site-specific nutrient management tool developed by International Rice Research Institute (IRRI), Philippines as shown in Fig. 2. Study compared the nutrient management among RCM with the blanket application of nutrients state fertiliser recommendation (SFR) and farmer's fertiliser practices (FFP) in eastern India due to its diverse soils, micronutrient deficiencies, management limitations and lower yields provide circumstances in which SSNM tool RCM offer significant benefits over farmer's fertiliser methods. The tool provides fertiliser recommendations determined by the local soil and crop conditions, climate and nutrient state (21). The grain yield was recorded maximum with RCM as compared to FFP (22).

Web tools based on SSNM are easily available and validated by the Indian Council for Agricultural Research (ICAR) institutes and universities, available for extension workers and farmer's (22). Rice crop manager uses information collected from the farmer on the name of variety and age of transplanted seedlings and then assessed information on growth and duration of the selected variety, which was stored in a database, to calculate a recommendation with days after transplanting for each N fertiliser application. A set of 12–15 questions asked by the web-based tool RCM, are relative to the history of the field, cultivation practice and weed management recommendations which provides the fertiliser recommendation for the crop field. The studies reveal that the RCM have an advantage to save the nitrogenous fertiliser over the blanket application by the farmer's. The improved performance of RCM can be attributed to enhanced N uptake resulting from increased availability of both macro and micronutrients (22–24). Two year average significant increase in grain yield of puddled transplanted rice by RCM (57.6 % and 39.7 %) compared to FFP and SFR, respectively. In Northern Philippines, RCM yielded significantly higher than FFP in 2011 and 2012 and was marginally significantly higher ($P < 0.10$) in 2013 as shown in Table 1.

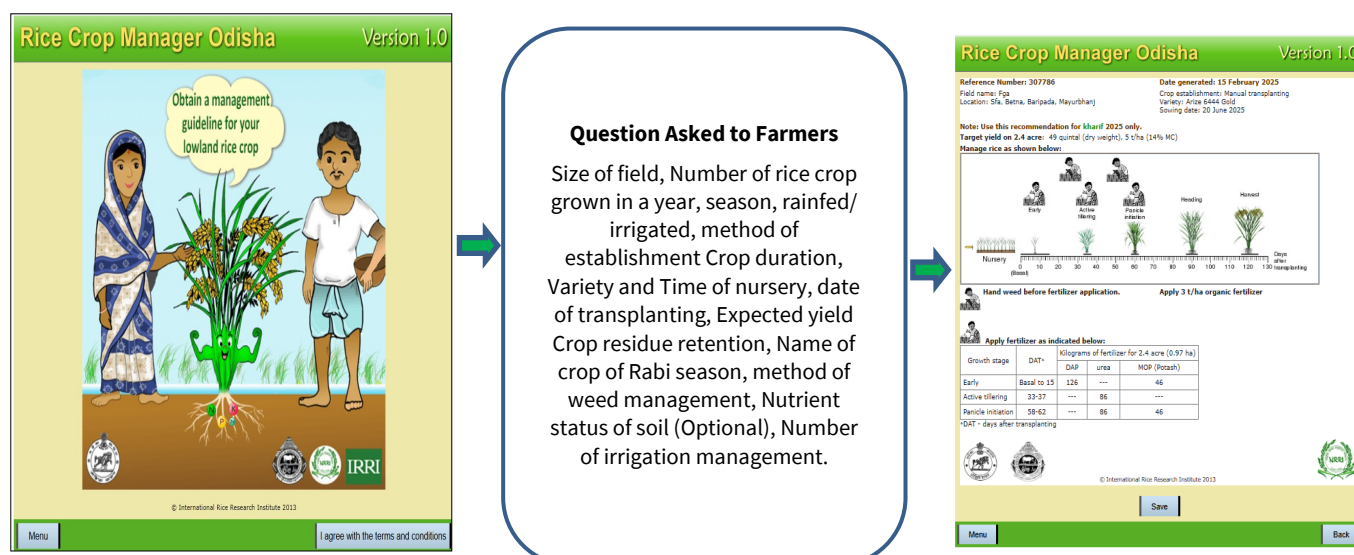


Fig. 2. Fertiliser recommendation suggested by the rice crop manager tool.

Table 1. Managing the nutrients by site-specific nutrient management tools for recommendation/reduction of greenhouse gases

Sl. No.	Tools/best treatment	Location	Recommendation/reduction of greenhouse gases	Reference
1	SSNM tool LCC	Punjab	Results of 350 on-farm experiment LCC based N management saved 9.4–54.2 kg N ha ⁻¹ .	(2)
2	SSNM tool LCC	Gurdaspur, Punjab	LCC based N management, from 9.4–54.2 kg N ha ⁻¹ , with an average of about 25 % less N fertiliser was used, without any reduction in yield as compared to the FP of applying blanket N at fixed time intervals. Application of N fertiliser when colour of the first fully expanded leaf was less than LCC shade 4, increased nitrogen use efficiency from 48–65 kg grain kg N ha ⁻¹ .	(16)
3	SSNM tool LCC and SPAD meter	Nadia, West Bengal,	Through the use of LCC and SPAD, 20–42.5 and 27.5–47.5 kg N ha ⁻¹ can be saved, respectively over that of the highest level of fixed-timing N applications.	(20)
4	SSNM tools (LCC/ RCM)	Banda University of Agriculture and Technology, Banda	Total CO ₂ reduction as compared to FFP and SFR. Two-year average significant increases grain yield of puddled transplanted rice by IRRI LLC (24.8 % and 9.1 %), PAU LCC (26.8 % and 10.8 %) and RCM (20.0 % and 4.9 %) over FFP and SFR, respectively. The two-year mean agronomic efficiency was found to be significantly better with International Rice Research Institute LCC (57.5 % and 39.6 %), Punjab Agriculture University LCC (52.1 % and 34.8 %) and RCM (57.6 % and 39.7 %) compared to FFP and SFR, respectively.	(23)
5	SSNM tool RCM	Odisha	Grain yield was consistently higher with fertilisation recommended by RCM than with FFP. Higher yield with RCM was attributed to a combination of applying more of the total fertiliser N at the critical growth stage of panicle initiation, applying more fertiliser N in kharif and applying zinc.	(22)
6	SSNM practice	International Rice Research Institute, Phillipines, Vietnam	Reduction in fertiliser use with SSNM average 10 % in Philippines and 14 % in Vietnam compared to FFP	(28)
7	SSNM Tool Nutrient expert	India	A 2.5 % reduction in global warming potential associated with reduced GHG emissions, increased rice yields and profit in India when nutrient expert was compared to FFP.	(29)
8	SSNM practice	One hundred and seventy-nine on-farm experiments with irrigated rice in China, India, Indonesia, Thailand, the Philippines and Vietnam.	Nitrogen applications split in SSNM were about 10 kg N ha ⁻¹ or 25 % less than in FFP.	(31)
9	SSNM package practice	Punjab province north western part of India.	The performance of SSNM was tested for two rice crops. Compared with the current FFP, average grain yield increased from 5.1–6.0 kg ha ⁻¹ , while plant N, P and K accumulations increased by 13–15 %. The (GRF) was about 14 % greater with SSNM than with FFP.	(33)
10	SSNM Tool LCC and SPAD	Village Tianmushan in Jinhua, China	Results of the comparison of SSNM with FFP in the double rice cropping system. Site-specific nutrient management increased rice yields by 0.4 t ha ⁻¹ , while fertiliser N was reduced by about 30 %	(35)
11	SSNM tool LCC	Seven field experiments with two fertilisation methods, SSNM and FFP, were conducted in 2004 and 2005 in the Heilongjiang Province, China	Compared with FFP, SSNM significantly decreased the average N rate by 33.8 %, as well as increased the average yield by 9.8 % in 2004 and 2005.	(36)
12	SSNM tool SPAD	Xushan Village, Hubei Province, China	The yield increase due to N application was 2.2–2.8 t ha ⁻¹ . SSNM reduced the N rate by 32 % compared with FFP	(38)
13	SSNM tool LCC	Cauvery Delta, Tamil Nadu	Leaf colour chart real time N management significantly increased grain yield (0.3–0.5 t ha ⁻¹) in both seasons within the old and new delta soils as compare to FFP.	(39)
14	SSNM tool NMR nutrient management tool	Cauvery Delta, Tamil Nadu	Grain yield was 0.6–0.7 kg ha ⁻¹ higher (P < 0.05) with NMR than FFP in two of the three seasons	(40)
15	SSNM tool RCM	Northern Philippines	Rice crop manager yielded significantly higher than FP in 2011 and 2012 and was marginally significantly higher (P < 0.10) in 2013	(41)
16	SSNM tool chlorophyll meter (SPAD)	Punjab Agriculture University, Ludhiana	Wheat yield increased by 20 % when 30 kg N ha ⁻¹ was applied at SPAD value of 42 at MT. Results show that plant need based N management through chlorophyll meter reduces N requirement of rice from 12.5–25 %, with no loss in yield.	(42)

16	SSNM tool chlorophyll meter (SPAD)	Punjab Agriculture University, Ludhiana	Wheat yield increased by 20 % when 30 kg N ha ⁻¹ was applied at SPAD value of 42 at MT. Results show that plant need based N management through chlorophyll meter reduces N requirement of rice from 12.5–25 %, with no loss in yield.	(42)
17	SSNM tool LCC	Punjab Agriculture University, Ludhiana	Fixed date adjustable dose N management through LCC produced yield levels higher than those achieved by applying blanket recommendation for N fertiliser and resulted in agronomic efficiency higher than 25 kg grain/kg N.	(43)
18	SSNM tool LCC	The six villages were Chandbill and Monoharpur in Meherpur District, Bangladesh	In rice, use of the LCC for N management without any other change in farmer's fertiliser and crop management increased mean grain yield by 0.3–0.4 kg ha ⁻¹ across villages and seasons.	(44)
19	SSNM practice	The experimental domain is located in the Jinhua-Quzhou Basin in central Zhejiang, near Jinhua City, China	Site-specific nutrient management performed equally better than FFP due to improved timing and splitting of N fertiliser which increased N recovery efficiency from 0.18 kg kg ⁻¹ N in FFP plots to 0.29 kg kg ⁻¹ N in SSNM plots. The agronomic N use efficiency (grain yield increase per kg fertiliser applied) was 80 % greater with SSNM than with FFP.	(45)
20	SSNM tool LCC	Punjab Agricultural University Research Farm at Ludhiana	Use of LCC with a critical shade 4 for N management in hybrid rice resulted in the application of 80–110 kg N ha ⁻¹ , thus effecting a saving of 10–40 kg N ha ⁻¹ compared to application 120 kg N ha ⁻¹ in three equal splits at fixed growth stages. Application of 120 kg N/ha increased the grain yield by 65.9, 64.6 and 54.4 % over no N control during 2006, 2007 and 2008 respectively.	(46)

Note: CO₂: carbon dioxide; FFP: farmer's fertiliser practices; FP: farmer's practice; GHG: greenhouse gases; GRF: gross return above fertiliser cost; K: potassium; LCC: leaf colour chart; N: nitrogen; P: phosphorus; RCM: rice crop manager; SFR: state fertiliser recommendation; SPAD: soil plant analyser development; SSNM: site-specific nutrient management; MT: maximum tillering; NMR: nutrient manager for rice.

Impact of site-specific nutrient management tools on greenhouse gas production, yields and efficiency indices

Greenhouse gas emission

The high requirement of water during cultivation of rice leads to a rise in temperature caused by the emission of GHGs such as carbon-dioxide (CO₂), N₂O and methane (CH₄) (25). Greenhouse gases from the rice fields can be mitigated through proper management of fertilisers, tillage practices and water. According to the past studies the total emission of GHG, CO₂ is found to be a lower contributor in omission plot as fertiliser was not added. Therefore, N is one of the most important factors of GHGs emission, the growing of crop without N leads to substantial yield penalty. Hence, application of N is imperative for getting better yield of rice. Nitrogen management through SSNM tools had lower CO₂ emission by 11.3 % than omission plot. Among the N management practices, the RCM had the lowest GHG emission. The CO₂ emission per kg of grain produced was highest in omission plots and lowest under N management using Punjab Agriculture University (PAU) LCC and IRRI LCC (23).

Management of fertilisers, weed, water and crop through SSNM tools have potential to obtain higher yields with increased N fertiliser use efficiency while maintaining low N₂O emissions (26). Compared with control treatment, the two-year average ammonia (NH₃), N₂O and nitric oxide (NO) emissions under LCC + urea (SSNM1) were reduced by 45.2 %, 19.2 % and 22.0 %, respectively. Furthermore, over two years, the average NH₃, N₂O and NO emissions under LCC + controlled release of urea (SSNM2) were 70.4 %, 34.4 % and 36.7 % lower, respectively, than those under control treatment (27). The reason is that N supply and crop demand are now better aligned. By preventing the overuse and inappropriate timing of fertiliser application, SSNM lowers the buildup of mineral N (ammonium ion (NH₄⁺) and nitrate ion (NO₃⁻)) in the soil floodwater system. Efficient application of N fertilisers decrease N₂O emission and global warming resulting in higher yields. The SSNM approach to fertiliser application doesn't enhance the N₂O emission per unit of grain yield (28). As same as nutrient expert, SSNM tool manage the

fertiliser application in rice and wheat crops. More than 80 % of farmer's have increased their yield and farm income by using the nutrient index tool. Nutrient index-based recommendation reduces the global warming potential (GWP) by about 2.5 % in rice crop and 12–20 % in wheat as compared to FFP (29).

Yield of rice

Grain yield increased with N application are varied depending on the treatment compared to the control. The yield target was determined by taking the omission plot technique from grain yield in N fertilised plot with limited nutrients in another plot. Nitrogen management through SSNM tools i.e. PAU LCC, IRRI LCC and RCM directly influences the yield of rice. Yield recorded with IRRI LCC and PAU LCC are statistically similar and highest grain yield as compared to FFP and RCM (23). Study on farmer's field, has been seen, SSNM practices produced 5 % higher yield than FFP. The reason behind it was associated with the reduction in insect and disease infestation and right application of optimal fertiliser dose at the right time improved lodging resistance (30). Improved N supply synchronisation with crop demand and the real time N application based on LCC enhances the N supply at the precise moment that the crop needs it the most, promoting N partitioning into yield components rather than excessive vegetative development that encourages pest outbreaks. Balanced N availability during critical development stages enhances photosynthesis, leaf area index, nutrient absorption and grain filling, which increases yield components while avoiding excessive vegetative biomass that might cause lodging (27). Site-specific nutrient management has potential to fulfill the yields gap and nutrient efficiency in irrigated rice. According to research findings the average yield under SSNM was recorded as 7 % increased yield as compared to FFP and profitability by 12 % (31). Number of trials conducted under site-specific approach for nutrient management in irrigated transplanted rice crops in Northwest India shows that field-specific management of macronutrients increases yields of rice crop by 12 % and profitability by 14 % as shown in Fig. 3 (32). Several studies reveals that higher yield was influenced by different SSNM tool as compared to FFP.

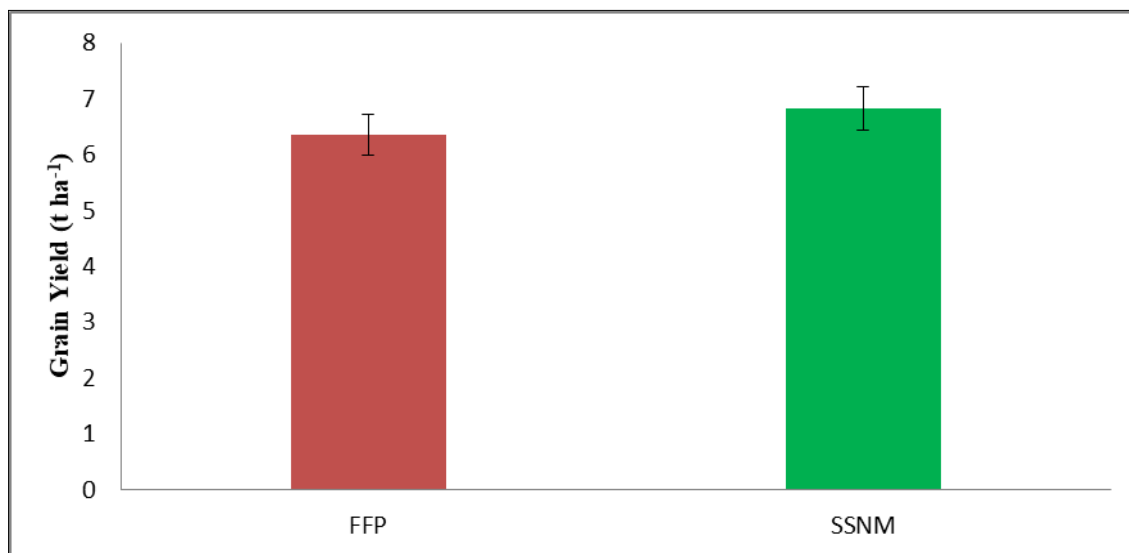


Fig. 3. Improvement in rice yield in different experimental findings.

Nutrient use efficiency

In this review study, different nutrient use efficiency like agronomic use efficiency, recovery efficiency and partial factor productivity of different research are used as comparison factor in variable treatment. Nutrient use efficiency is a term used for a ratio of an output (biological yield or economic yield) to the input (nutrient supply). The total aboveground plant biomass (plant dry matter) and total plant N differ from grain yield and grain N, which are components of the economic yield (6). The main objective of nutrient use efficiency (NUE) is to increase the crop production while minimising the nutrient loss from the field. Nutrient use efficiency can be increased by better agricultural management practices.

Agronomic efficiency: Agronomic efficiency (AE) is calculated in units of yield increase per unit of nutrient applied. It directly reflects the production of a crop when fertiliser is applied and relates directly to economic return. On the basis of LCC the critical value based N management on farm experiment in Punjab revealed that the N saving was 9.4–54.2 kg N ha⁻¹, with an average of about 25 % without any decrease in yield as compared to FFP of applied blanket N at fixed time intervals. By using the LCC, application of N fertiliser using LCC increased agronomic N use efficiency from 48 kg grain kg⁻¹ N to 65 kg grain kg⁻¹ N (2). Real time N management through SSNM tools increases the AE in rice as it improves the growth parameters such as

number of tillers, yield attributes and yield (33). Different research paper entries shows that, after the analysis 23 % improvement in AE exhibits better utilisation of N by the crop as shown in Fig. 4 (30). The performance of SSNM across 107 sites in China, evaluated over 10 seasons, showed that the average N applied through SSNM (133 kg N ha⁻¹) was lower than that under FFP (195 kg N ha⁻¹). Experiments showed that an increase in yield of 5 % with SSNM, profit was observed with AE which is 61 % greater as compared to FFP (34). Nutrient management through SSNM was evaluated across 21 farm sites over four successive rice crops to assess its performance under field conditions. The study demonstrated that SSNM significantly improved fertiliser efficiency, with agronomic nitrogen use efficiency (AE_N) being 80 % higher than under FFP. This increase was reflected in greater grain yield per kg of fertiliser applied, indicating superior synchronisation of N supply with crop demand under SSNM (35). Site-specific nutrient management tool LCC value is used in experiments of two fertilisation methods, SSNM and FFP were conducted considerably improved average tillers, leaf area index, dry matter and N uptake. Recovery efficiency and agronomic efficiency increased 38.5–133.4 % respectively, over the FFP. Site-specific nutrient management has likely increased yield and nutrient efficiency in irrigated and transplanted rice (36).

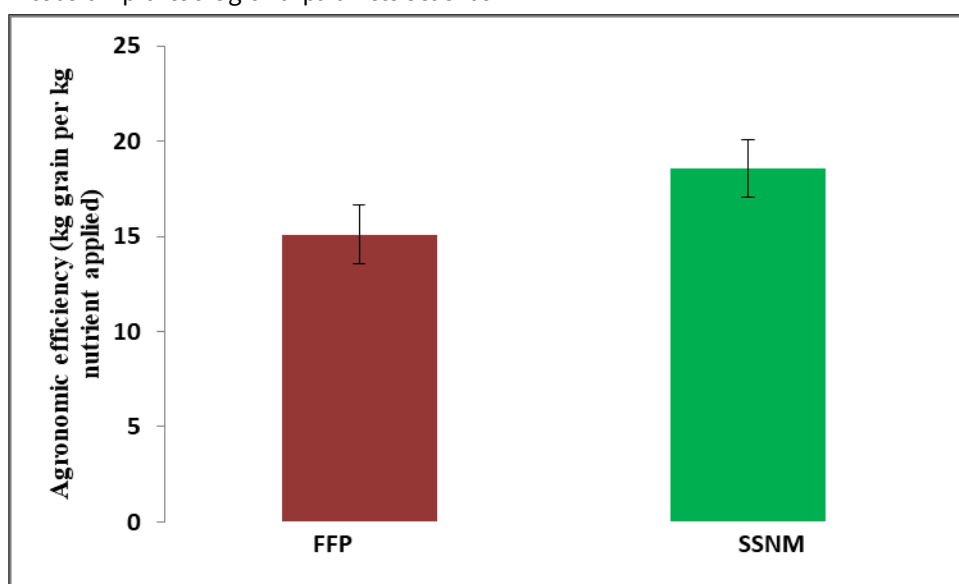


Fig. 4. Agronomic efficiency in farmer's fertiliser practice and site-specific nutrient management.

Partial factor productivity: Partial factor productivity (PFP_N) is the easiest and simplest production efficiency in determining the units of crop yield per unit of applied nutrient and to maintain the data of inputs and yield. The total productivity of applied N is measured by the PFP_N . The impact of N fertiliser, the contribution of indigenous soil N, the impact of management techniques and the influence of weather and crop type are all reflected in it. It is simple to compute from farmer's field data and it only needs two data points yield and N applied. It does not need a control plot, lab analysis and plant N uptake measurements. Agronomic efficiency requires fertilised plot and an unfertilised control plot. Recovery efficiency (RE) requires plant tissue sampling, N uptake in grain and in straw and control treatment, but the PFP_N requires only fertilised yield and nutrient applied in crop field. With increase in PFP_N (21 % in the Philippines and 26 % in Vietnam), SSNM reduces the N fertiliser application and increases yield. An increase in PFP_N could reasonably be associated with increased plant use of N fertiliser and reduced loss of N fertiliser. The simulated reduction in N fertiliser loss with SSNM is consistent with the measurements for N fertiliser use efficiency as PFP_N (28). Same as with PFP_N , the SSNM tool practices achieve 15.54 % higher production as compared to FFP as shown in Fig. 5.

Recovery efficiency: One of the complex forms of nutrient use efficiency expression is RE, defined as the dissimilarity in nutrient uptake in above ground parts of the plant between fertilised and unfertilised crop comparative to the amount of nutrient applied. Nutrient applied in soil and half of the nutrient especially N is not assimilated by the crops (11). Simulated N loss for FFP (55 %) is somewhat higher than the direct-seeded rice (26 %) and transplanted rice (37 %) losses of N fertiliser measured for farmer's fertiliser split in the above mentioned ^{15}N balance research (37). Enhanced recovery efficiency was associated with N application managed through SSNM tools. Higher recovery efficiency was recorded with IRRI LCC and RCM by using SSNM tools technique to manage the N fertiliser doses in transplanted rice over the blanket application of N fertiliser (23). Slight yield increase was achieved when total N rate was reduced from 195–133 kg ha⁻¹ and 32 % saving of N fertiliser over farmer's N fertiliser practice. It was noticed that N response of SSNM was 28 % higher than that of farmer's N fertiliser practice as shown in Fig. 6. Improved rice crop lodging resistance and reduced insect and disease damage were related to the yield increase at the best N inputs. The primary cause of the rice crops low fertiliser N usage efficiency is that the majority of rice farmer's utilise

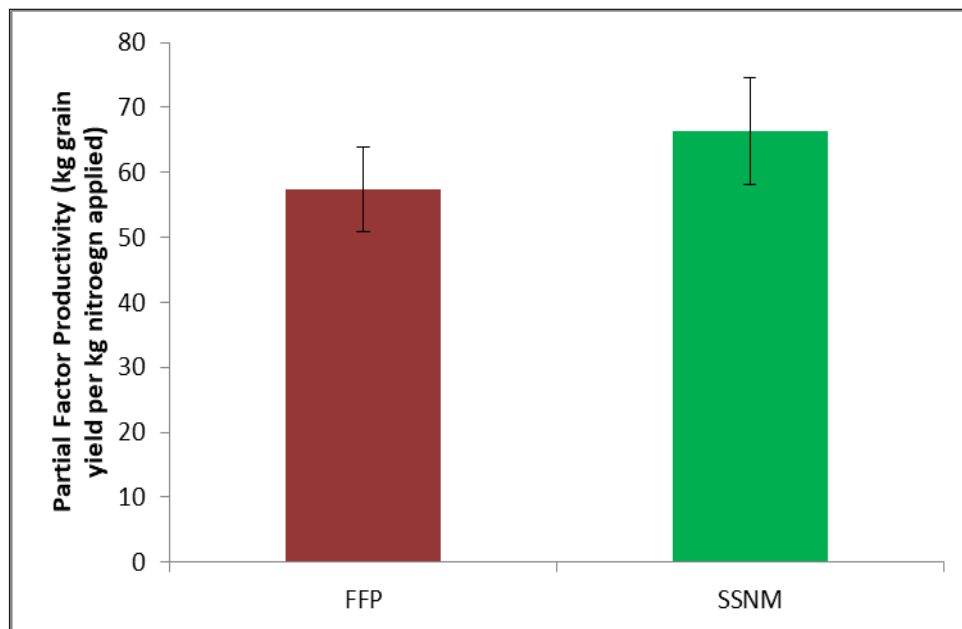


Fig. 5. Partial factor productivity in farmer's fertiliser practice and site-specific nutrient management.

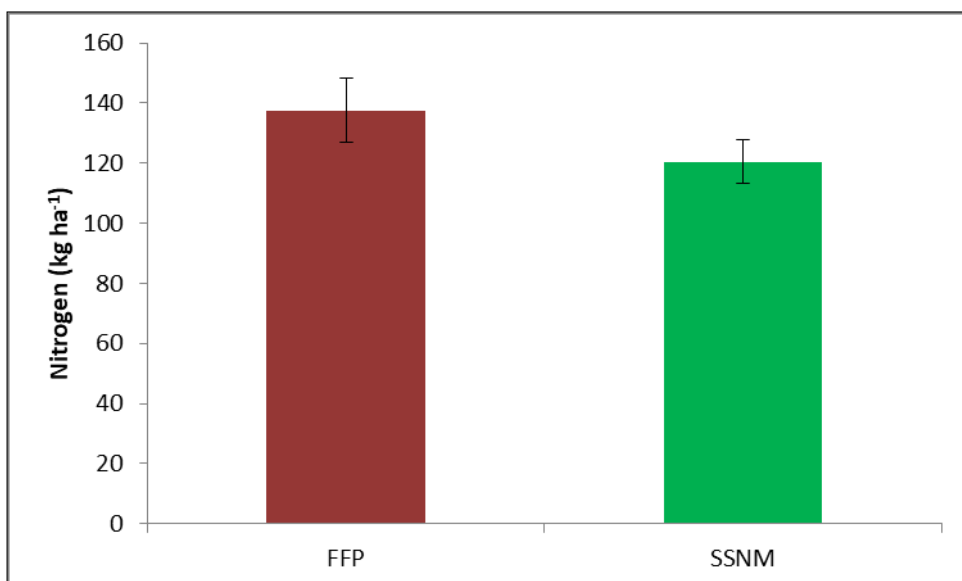


Fig. 6. Recovery efficiency in farmer's fertiliser practice and site-specific nutrient management.

excessive amounts of N fertiliser, particularly during the early vegetative stage. Researchers discovered that irrigated rice fields have 50 % more native N supply capacity. Site-specific nutrient management is an established technology for enhancing the grain yield and N fertiliser utilisation efficiency of rice plants (30). The different studies shows that the SSNM tools certify the higher absorption of applied N with saving the amount of fertiliser applied in crop field and reduced the amount of fertiliser cost.

Conclusion

The research papers highlighted the importance of site-specific nutrient management (SSNM) in sustainability of the rice production in South Asia. It is important to ensure the effective management of nitrogen (N). A reduction in the dose of N is possible through proper tools without yield penalty. The greenhouse gases (GHGs) footprint may also reduce from the rice field through the SSNM tools. Compared to the farmer fertiliser practice (FFP), SSNM in rice farming systems boosts rice production, profitability and N utilisation efficiency while lowering N losses. The farmer's practice resulted in 23 % lower agronomic nitrogen use efficiency (AE_N) and 15.54 % lower partial factor productivity (PPF_N) compared to SSNM despite utilising 14.3 % less N fertiliser. Balanced fertilisation combined with improved distribution and increased N fertiliser splits throughout the growing season are the main reasons for SSNM's superior performance over the farmer practice. Even though are several benefits for SSNM tools, the popularity of SSNM tools among the farming community is very less.

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Authors' contributions

ASS and JP conceived the idea and edited the manuscript. ASS and AM¹ wrote the review manuscript. KMT and AR assembled the data, AY and AM² designed the graph. All authors read, revised and approved of the manuscript. [AM¹ stands for Amit Mishra, AM² stands for Abhishek Maurya].

Compliance with ethical standards

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